JEE–MAIN EXAMINATION – APRIL 2026

(HELD ON MONDAY 06th APRIL 2026) **TIME : 3:00 PM TO 6:00 PM**

PHYSICS

TEST PAPER WITH SOLUTION

SECTION-A

26. The percentage error in the calculated volume of a sphere, if there is 2% error in its diameter measurement, is ____.

(1) 1 (2) 2

(3) 6 (4) 8

Sol. $v = \frac{4}{3}\pi R^3$

$\frac{\Delta v}{v} = \frac{3\Delta R}{R}$

$= 6\ \%$

27. Match List-I with List-II.

	List-I		List-II
A.	Boltzmann constant	I.	$[M^{-1}L^{3}T^{-2}]$
B.	Stefan's constant	II.	$[M\,L^{2}T^{-1}]$
C.	Planck's constant	III.	$[ML^{2}T^{-2}K^{-1}]$
D.	Gravitational constant	IV.	$[ML^{0}T^{-3}K^{-4}]$

(1) A-I, B-II,C-III, D-IV

(2) A-IV, B-III,C-II, D-I

(3) A-III, B-IV,C-II, D-I

(4) A-II, B-I,C-IV, D-III

Sol. (A) $\rightarrow ML^{2}T^{-2}K^{-1}$ (III)

(B) $\rightarrow MT^{-3}K^{-4}$ (IV)

(C) $\rightarrow ML^{2}T^{-1}$ (II)

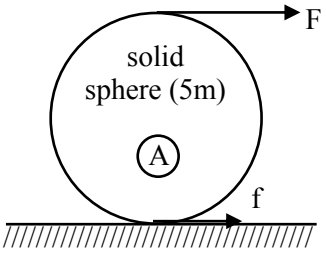
(D) $\rightarrow M^{-1}L^{3}T^{-2}$ (IV)

28. A solid sphere (A) of mass, 5m and spherical shell (B) of mass m, both having same radius, are placed on a rough surface. When a force of same magnitude is applied tangentially at the highest points of A and B, they start rolling without slipping with an acceleration of a_A and a_B, respectively. The ratio of a_A and a_B is _____.

(1) 5 : 21 (2) 6 : 10

(3) 21 : 25 (4) 1 : 5

Sol.



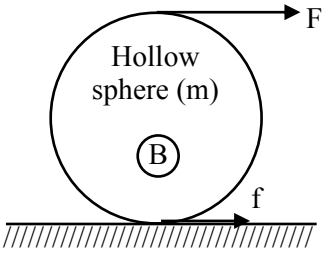
$F + f = 5m\,a_A$

$F(R) - f(R) = \frac{2}{5}(5m)R^2.\alpha$

$\alpha = R\alpha$

$2F = 7m\,a_A$

$a_A = \frac{2F}{7m}$



$F + f = ma_B$

$F(R) - f(R) = \frac{2}{3}mR^2 \times \alpha$

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$a = R\alpha$

$2F = \frac{5ma_B}{3}$

$\frac{a_A}{a_B} = \frac{2F}{7m} \times \frac{5m}{6F} = \frac{5}{21}$

29. A body of mass 1kg moves along a straight line with a velocity $v = 2x^2$. The work done by the body during displacement from x = 0 to 5m is ____J.

(1) 0 (2) 250

(3) 1250 (4) 1000

Sol. $w.\Delta = \Delta K.E$

$= \frac{1}{2}m\left(v_f^2 - v_i^2\right)$

$= \frac{1}{2} \times 1\left[\left(2(5)^2\right)^2 - \left(2(0)\right)^2\right]^0$

$= \frac{1}{2} \times 50 \times 50 = 1250J$

30. A cylinder with adiabatic walls is closed at both ends and is divided into two compartments by a frictionless adiabatic piston. Ideal gas is filled in both (left and right) the compartments at same P,V, T. Heating is started from left side until pressure changes to $\frac{27P}{8}$. If initial volume of each compartment was 9 litres then the final volume in right-hand side compartment is ____ litres. (for this ideal gas $C_p/C_v = 1.5$)

(1) 3 (2) 4

(3) 14 (4) 9

Sol. $P_{Left} = P_{Right}$ (final)

for right chamber

$P_iV_i^\gamma = P_fV_f^\gamma$

$P(9)^{3/2} = \frac{275}{8} \times V_f^{3/2}$

$27 = \frac{27}{8}V_f^{3/2}$

$\therefore V_f = 4$ litres

31. For an electromagnetic wave propagating through vacuum, $\vec{k}$, $\vec{E}$ and ω represent propagation vector, electric field and angular frequency, respectively. The magnetic field associated with this wave is represented by :

(1) $\frac{\vec{E} \times \vec{k}}{\omega}$ (2) $\frac{\vec{k} \times \vec{E}}{\omega}$

(3) $\omega(\vec{E} \times \vec{k})$ (4) $\omega(\vec{k} \times \vec{E})$

Sol. $\vec{B} = \frac{1}{\omega}\left(\vec{k} \times \vec{E}\right)$

32. Two identical bodies A and B of equal masses have initial velocities $\vec{v}_1 = 4\hat{i}\,m/s$ and $\vec{v}_2 = 4\hat{j}\,m/s$ respectively. The body A has acceleration $\vec{a}_1 = 6\hat{i} + 6\hat{j}\,m/s^2$ while the acceleration of the other body B is zero. The centre of mass of the two bodies moves in _____path.

(1) circular (2) parabolic

(3) straight line (4) elliptical

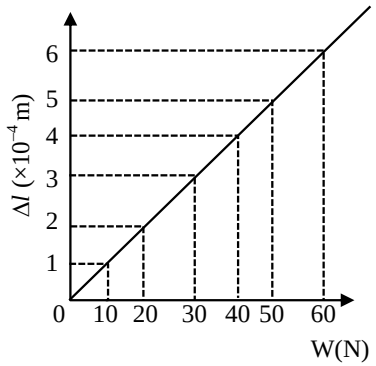
Sol. $\vec{v}_{cm} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2} = \frac{m(4\hat{i}) + m(4\hat{j})}{2m} = 2\hat{i} + 2\hat{j}$

$\vec{a}_{cm} = \frac{m_1\vec{a}_1 + m_2\vec{a}_2}{m_1 + m_2} = \frac{m(6\hat{i} + 6\hat{j}) + m(0)}{2m} = 3\hat{i} + 3\hat{j}$ -

$\vec{v}_{cm} \parallel \vec{a}_{cm}$

so, path is straight line

33. Figure represents the extension (Δl) of a wire of length 1 meter, suspended from the ceiling of the room at one end with load W connected to the other end. If the cross- sectional area of the wire is 10^{-5} m^2 then the Young's modulus of the wire is ___N/m^2.



(1) 1.0×10^{11} (2) 2.0×10^{10}

(3) 1.0×10^{10} (4) 2.0×10^{11}

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Sol. $\frac{\Delta\ell}{w}=\tan\theta=\frac{\Delta y}{\Delta x}=\frac{1\times10^{-4}}{10}=10^{-5}$

$\frac{w}{\Delta\ell}=10^{5}$

$\Rightarrow\ y=\frac{F/A}{\Delta\ell/\ell}=\frac{F\ell}{A\Delta\ell}=\left(\frac{w}{\Delta\ell}\right)\frac{\ell}{A}$

$\Rightarrow\ y=\frac{10^{5}\times1}{10^{-5}}=10^{10}\,N/m^{2}$

34. A cylindrical vessel of 40 cm radius is completely filled with water and its capacity is 528 dm^3 (dm : decimeter). The vessel is placed on a solid block of exactly same height as vessel. If a small hole is made at 70 cm below the top of water level, then horizontal range of water falling on the ground in the beginning is ____cm.

(1) $120\sqrt{2}$ (2) $140\sqrt{2}$

(3) $140\sqrt{3}$ (4) $120\sqrt{3}$

Sol. $v=\pi r^2h\Rightarrow528\times10^{-3}=\pi(0.4)^2\,h$

$h=\frac{528\times10^{-3}\times7}{(0.4)^2\times22}=1.05m=105cm$

$\rightarrow\ t=\sqrt{\frac{2H}{g}}=\sqrt{\frac{2\times(105+35)\times10^{-2}}{10}}$

$=\sqrt{\frac{2(140)\times10^{-2}}{10}}\ \sqrt{28}\times10^{-1}sec$

$R=ut=\sqrt{2\times10\times70\times10^{-2}}\times\sqrt{28}\times10^{-1}$

$=\left(\sqrt{14}\right)\left(\sqrt{28}\right)\times10^{-1}=14\sqrt{2}\times10^{-1}m$

$R=140\sqrt{2}$ cm

35. If 2 mole of an ideal monoatomic gas at temperature T, is maxed with 6 mole of another ideal monoatomic gas at temperature 2T then the temperature of mixture is :

(1) $\frac{5}{2}T$ (2) $\frac{5}{4}T$

(3) $\frac{7}{2}T$ (4) $\frac{7}{4}T$

Sol. $T_{mix}=\frac{n_1C_1T_1+n_2C_2T_2}{n_1C_1+n_2C_2}$

$=\frac{\left(2\times\frac{3}{2}R\times T\right)+\left(6\times\frac{3}{2}R\times2T\right)}{\left(2\times\frac{3}{2}R\right)+\left(6\times\frac{3}{2}R\right)}$

$T_{mix}=\frac{2T+12T}{8}=\frac{7}{4}T$

36. A spring stretches by 2 mm when it is loaded with a mass of 200 g. From equilibrium position the mass is further pulled down by 2 mm and released. The frequency associated with the system and maximum energy in the spring are _____ Hz and _____ J, respectively. (Take g = 10 m/s^2)

(1) $\frac{5\sqrt{50}}{\pi}$ and 8×10^{-3}

(2) $\frac{5\sqrt{50}}{\pi}$ and 8

(2) $10\sqrt{50}$ and 2×10^{-3}

(4) $\frac{5\sqrt{50}}{\pi}$ and 16×10^{-3}

Sol. $kx=0.2g;$ $\qquad f=\frac{1}{2\pi}\sqrt{\frac{k}{m}}$

$k\times2\times10^{-3}=2$ $\qquad f=\frac{1}{2\pi}\sqrt{\frac{1000}{0.2}}$

$k=1000\ N/m$

$=\frac{1}{2\pi}\sqrt{5000}=\frac{5}{\pi}\sqrt{50}$

$E_{max}=\frac{1}{2}k\ x_{max}^2$

$\frac{1}{2}\times1000\times\left(4\times10^{-3}\right)^2=8\times10^{-3}$ Joule

37. The electric potential as a function of x, y is given by $V=5(x^2-y^2)V$. The electric field at a point (2, 3) m is _____ V/m.

(1) $(-20\hat{i}+30\hat{j})$ (2) $(20\hat{i}-30\hat{j})$

(2) $(20\hat{i}+45\hat{j})$ (4) $(-4\hat{i}+6\hat{j})$

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Sol. $V = 5(x^2 - y^2)V$

$$\varepsilon_x = \frac{-dv}{dx} = -10x\hat{i}$$

$$\varepsilon_y = \frac{-dv}{dx} = +10y\hat{i} \Rightarrow \varepsilon(2,3) = -20\hat{i} + 30\hat{j}$$

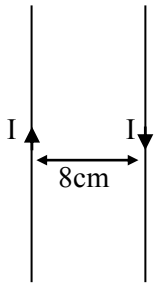
38. A current of 30 A each flows in opposite directions in two conducting wires, placed parallel to each other at a distance of 8 cm. The magnetic field at the mid point between the two wires is _____ μT.

$\left(\frac{\mu_0}{4\pi} = 10^{-7}\ N/A^2\right)$

(1) 30 (2) 300

(2) 150 (4) 0.1

Sol.



$$B = \frac{\mu_0 I}{2\pi(4)\times 10^{-2}} \times 2$$

$= 10^{-7} \times 30 \times 10^2$

$= 30 \times 10^{-5}$ T

$= 300\ \mu T$

39. A square loop of side 2cm is placed in a time varying magnetic field with magnitude as B = 0.4 sin(300t) Tesla. The normal to the plane of loop makes an angle of 60° with the field. The maximum induced emf produced in the loop is ____ mV.

(1) 12 (2) 18

(3) 21 (4) 24

Sol. $B = 0.4\sin 300t \qquad \theta = 60°$

$A = (0.02)^2 = 4 \times 10^{-4}\ m^2$

$\phi = BA\cos\theta$

$\phi = 0.4 \sin(300t) A\cos\theta$

$$emf = \frac{d\phi}{dt} = 0.4\ A\cos\theta\cos(300t) \times 300$$

$$\text{max emf} = 0.4 \times 4 \times 10^{-4} \times 300 \times \frac{1}{2}$$

$= 240 \times 10^{-4}$ V

$= 24$ mV

40. A sphere of capacitance 100 pF is charged to a potential of 100V. Another identical undercharged metal sphere is brought in contact with the charged sphere, then the change in the total energy stored on these spheres, when they touch is $\alpha \times 10^{-7}$ J. The value of α is _____. (combined capacitance of spheres is 200 pF).

(1) 5 (2) $\frac{5}{2}$

(3) $\frac{7}{2}$ (4) $\frac{9}{2}$

Sol. $Q_1 = CV = 100 \times 100\ pC = 10^4\ pC$

$$V_1 = V_2 = \frac{V}{2} = 50$$

$$\varepsilon_i = \frac{1}{2}\times 100 \times (100)^2\ pJ = \varepsilon$$

$$\varepsilon_f = 2 \times \frac{1}{2} \times 100 \times (50)^2 = \frac{\varepsilon}{2}$$

$$\Delta\varepsilon = \frac{\varepsilon}{2} = 100 \times (50)^2\ pJ$$

$= 25 \times 10^4 \times 10^{-12} = 2.5 \times 10^{-7}$

41. The energy released if hydrogen atoms are combined to form $\frac{4}{2}He$ is_____.

(Take binding energies per nucleon of ^2_1H and ^4_2He as 1.1 meV and 7.2 MeV, respectively).

(1) 6.1 (2) 24.4

(3) 26.6 (4) 5

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Sol. ${}_1^2H + {}_1^2H \rightarrow {}_2^4He$

$Q = BE\ (RHS) - BE\ (LHS)$

$= 4 \times 7.2 - 4 \times 1.1$

$= 4 \times 6.1$

$Q = 24.4\ MeV$

42. Angle of minimum deviation is equal to the half of the angle of prism is an equilateral prism. The refreactive index of the prism is _____.

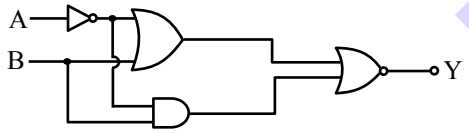
(1) 1.5 (2) $\sqrt{3}$

(3) $\sqrt{2}$ (4) 1.65

Sol. $\mu_p = \mu_s \dfrac{\sin\left(\dfrac{A+\delta_{min}}{2}\right)}{\sin\left(\dfrac{A}{2}\right)} = \dfrac{\sin\dfrac{3A}{4}}{\sin\dfrac{A}{2}}$

$\mu_p = \dfrac{\dfrac{1}{\sqrt{2}}}{\dfrac{1}{2}} = \sqrt{2} = \sqrt{2}$

43. Refer to the logic circuit given below. For two inputs (A = 1, B = 1) and (A = 0, B = 1), output (Y) will be _____.

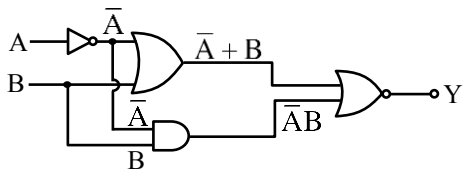


(1) 1, 0 respectively (2) 0,1 rsepectively

(3) 0,0 respectively (4) 1,1 respectively

(5) 0.0

Sol.



$y = \overline{\bar{A} + B + \bar{A}B}$

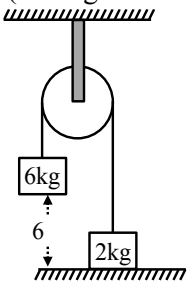
$= \overline{\bar{A}+B} = \bar{A} \cdot \bar{B}$

$= A\bar{B}$

for A = 1, B = 1, y = 0

for A = 0, B = 1, y = 0

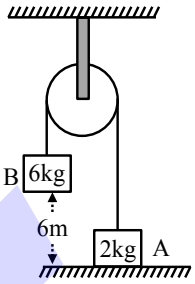
44. The velocity at which 6 kg mass (shown in figure) strikes the ground when it is released from a height of 6m above the ground is _____ m/s. Assume pulley is massless and string is light and inextensible (Take g = 10 m/s^2).



(1) 7.74 (2) 7.20

(3) 6.55 (4) 4.50

Sol.



$W_G = \Delta K$

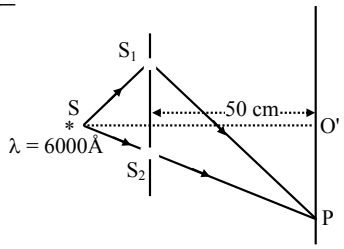
$m_B gh - m_A gh = \dfrac{1}{2}(m_A + m_B)V^2$

$(6-2) \times 10 \times 6 = \dfrac{1}{2}(8)v^2$

$v = \sqrt{60}\ m/s$

$v = 7.746\ m/s$

45. In a Young double slit experiment, the wavelength of incident of incident light is 6000 Å, the separation between slits S_1 and 5 cm the distance between slits plane and screen is 50 cm, as shown in the figure below. If the resultant intensity at P is equal to the intensity due to individual slits, the path different between interfering waves is _____Å.



(1) 4000 (2) 3000

(3) 2000 (4) 1000

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Sol. $I_P = I_0 = 4I_0 \cos^2 \frac{\Delta\phi}{2}$

$\cos \frac{\Delta\phi}{2} = \frac{1}{2}$

$\frac{\Delta\phi}{2} = \frac{\pi}{3}$

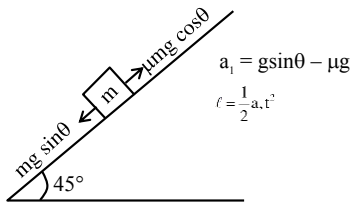
$\frac{2\pi}{\lambda} = \Delta x = \Delta\phi = \frac{2\pi}{3}$

$\Delta x = \frac{\lambda}{3} = 2000 A°$

SECTION-B

46. A block takes t time to slide down a plane inclined at 45° to the horizontal. If the surface is made smooth (frictionless), the block takes times $\frac{t}{2}$ to slide down the plane. The coeffcient of friction between the block and the inclined plane is $\left(\frac{\alpha}{100}\right)$. The value of α is ____.

Sol.



for frictionless surface $a_2 = g \sin\theta$

$\ell = \frac{1}{2} a_2 \frac{t^2}{4}$

$a, t^2 = a_2 \frac{t^2}{4}$

$g(\sin 45 - \mu \cos 45°) = \frac{g \sin 45}{4}$

$1 - \mu = \frac{1}{4}$

$\mu = \frac{3}{4} = \frac{\alpha}{100}$

$\alpha = 75$

47. The de Broglie weave length for an electron accelerated through the potential difference V_1 volt is λ_1. When the potential difference is changed to V_2 volt, the associated de Broglie wavelength is increased by 50%. If $(V_1/V_2) = (9/\alpha)$, then the value of α is _____.

Sol. $\lambda_1 = \frac{h}{mv} = \frac{h}{\sqrt{2meV_1}} = \lambda$

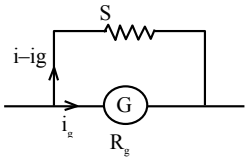
$\lambda_2 = \frac{h}{\sqrt{2meV_2}} = \frac{3\lambda}{2} = \frac{3}{2} \frac{h}{\sqrt{2meV_1}}$

$V_2 = \frac{4}{9} . V_1$

$\frac{V_1}{V_2} = \frac{9}{4} \Rightarrow \alpha = 4$

48. A moving coil of a golvanometer when shunted with 2Ω resistance gives a full scale deflection for a current of 500 mA. When a resistance of 470Ω is connected in series it gives a full scale deflection for 10V potential applied on it. The value of resistance of galvanometer coil is _____Ω.

Sol. $i_g R_g = 2\left(\frac{1}{2} - ig\right)$



$i_g = \frac{1}{R_g + 2}$

$10 = (R_g + 470) i_g$

$10 = \frac{R_g + 470}{R_g + 2}$

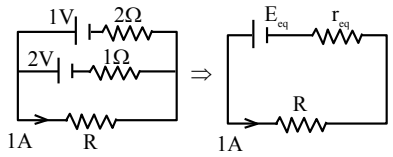
$10R_g + 20 = R_g + 470$

$R_g = 50\Omega$

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49. Two cells of emfs 1V and 2V and internal reseistance 2Ω and 1Ω, respectively connected in parallel, gave current of 1A through an external resistance. If the polarity of one cell is reversed, then value of current through the external resistance will be $\frac{\alpha}{5}$A. The value of α is _____.

Sol.



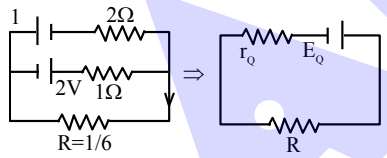
$$E_{eq} = \frac{\frac{1}{2}+\frac{2}{1}}{\frac{1}{2}+\frac{1}{1}} = \frac{\frac{5}{2}}{\frac{3}{2}} = \frac{5}{3}$$

$$r_Q = \frac{3}{2}$$

$$i = \frac{E_{eq}}{r_{eq}+R} = \frac{\frac{5}{3}}{\frac{3}{2}+R} = 1$$

$$\frac{5}{3} = \frac{3}{2} + R$$

$$R = \frac{1}{6}\Omega$$



$$r_{eq} = \frac{3}{2}$$

$$E_{eq} = \frac{\frac{2}{1}-\frac{1}{2}}{\frac{1}{1}+\frac{1}{2}} = 1$$

$$i = \frac{E_{eq}}{r_{eq}+R} = \frac{1}{\frac{3}{2}+\frac{1}{6}} = \frac{3}{5} \quad \Rightarrow \alpha = 3$$

50. A concave mirror of focal length 10 cm froms an image which is double the size of object when the object is placed at two different positions. The distance between the two positions of the object is ____ cm.

Sol. **Case I** $v = 2u$

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

$$\frac{1}{-2u}+\frac{1}{-u}=\frac{1}{-f}$$

$$\frac{3}{2u}=\frac{1}{f}$$

$$u_1 = \frac{3f}{2}$$

Case 2 $v = -2u$.

$$\frac{1}{+2u}+\frac{1}{-u}=\frac{1}{-f}$$

$$\frac{-1}{2u}=\frac{1}{-f}$$

$$u_2 = \frac{f}{2}$$

$$\Delta u = f = 10 \text{ cm}$$

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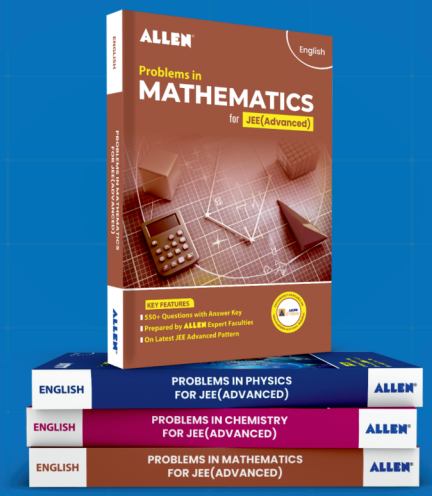


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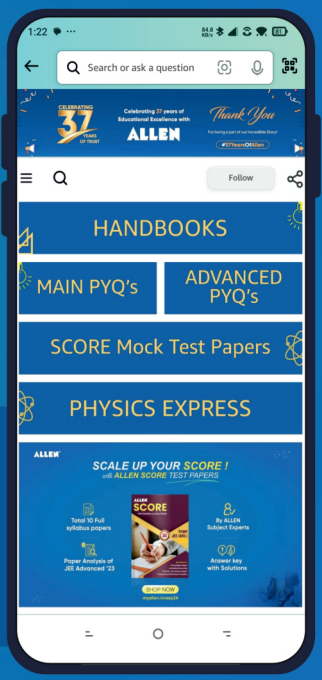
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